

Customer Churn Insights

Streamlit web dashboard for churn prediction, model comparison, explainability, and retention recommendations.

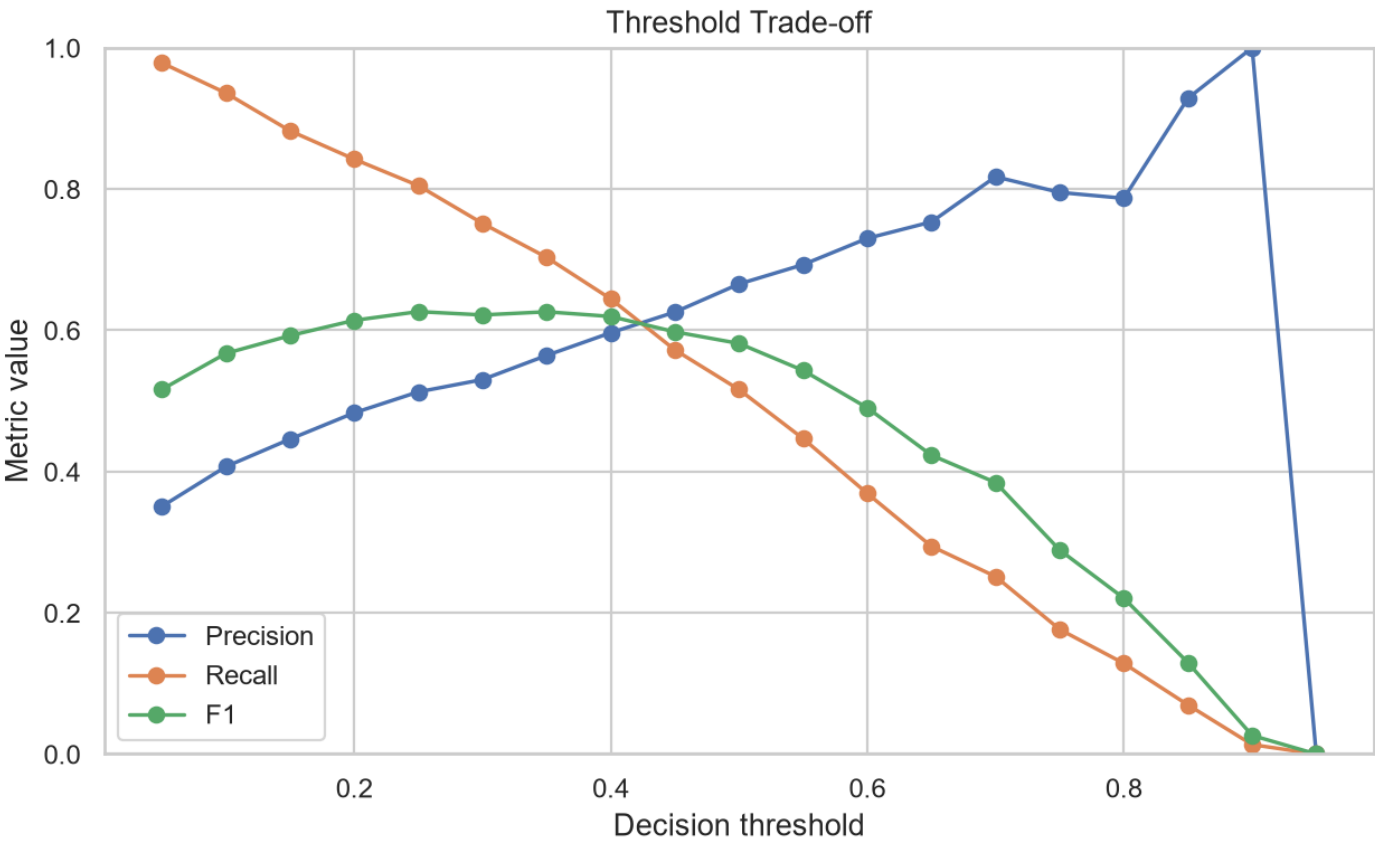
Customers	Churn rate	Best model	Best ROC-AUC
7,043	26.5%	Gradient Boosting	0.843
Prediction	Model Performance	Threshold Tuning	Business Insights

Threshold Tuning

The default classification threshold is 0.50, but churn campaigns often need a lower threshold to catch more at-risk customers.

Best F1 threshold	Precision	Recall	Predicted churn rate
0.25	0.513	0.805	41.7%

	threshold	precision	recall	f1	predicted_churn_rate	
4		0.25	0.513	0.805	0.626	41.7%
6		0.35	0.564	0.703	0.626	33.1%
5		0.30	0.530	0.751	0.622	37.6%
7		0.40	0.597	0.644	0.620	28.7%
3		0.20	0.483	0.842	0.614	46.3%
8		0.45	0.626	0.572	0.598	24.3%
2		0.15	0.446	0.882	0.592	52.5%
9		0.50	0.666	0.516	0.581	20.6%
1		0.10	0.407	0.936	0.567	61.0%
10		0.55	0.693	0.447	0.543	17.1%



Precision, recall, and F1 across decision thresholds